Not the common cause for this time! No sign of causal inference in audiovisual duration integration

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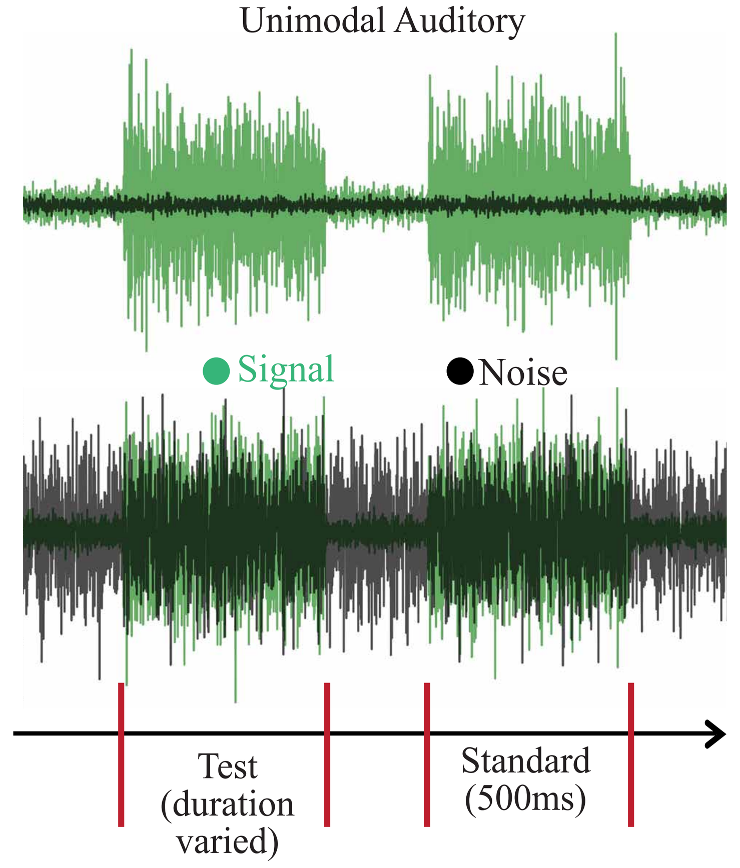
Introduction

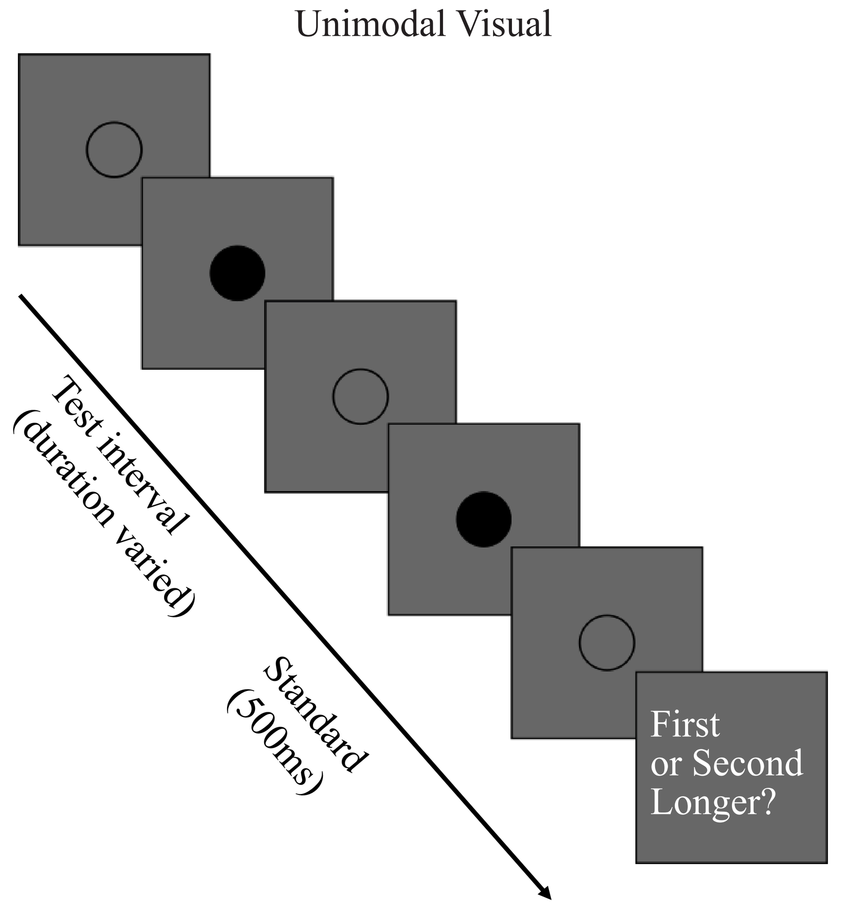
Audiovisual duration cues integrate optimally under small conflicts [1].

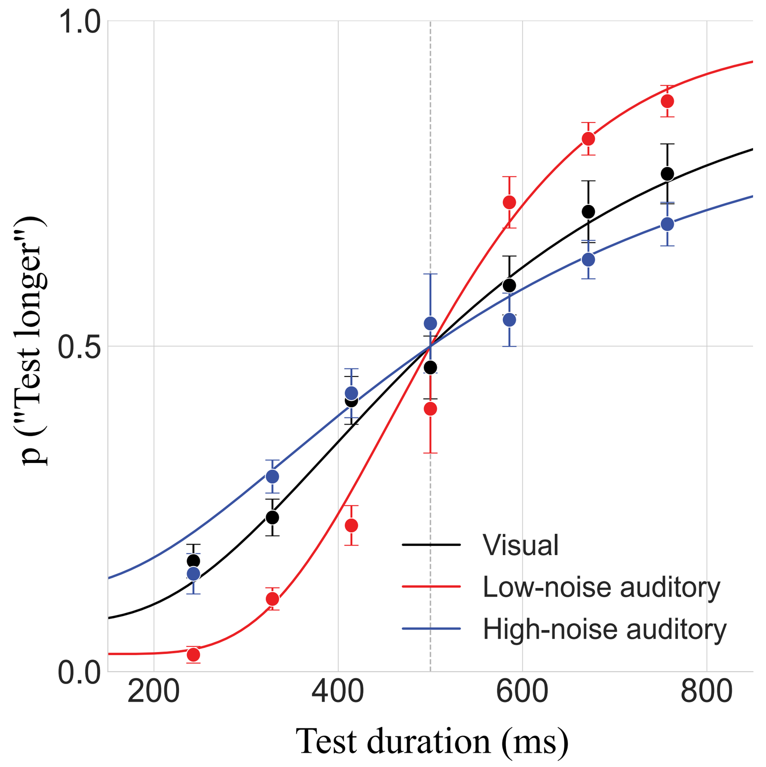
Does causal inference break integration for large audiovisual duration conflicts as with spatial localization [2]?

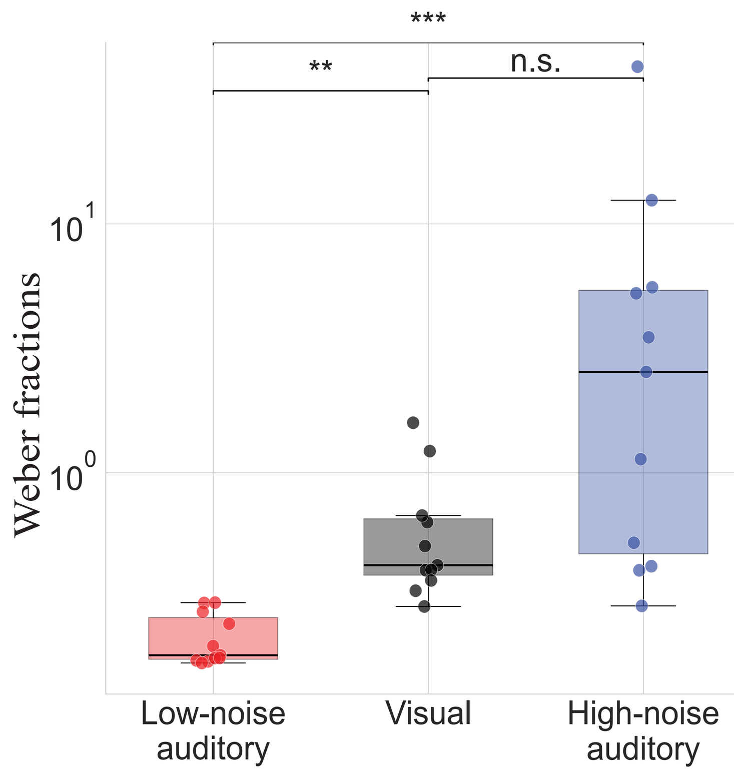
- Does the brain fuse, segregate, or switch between AV duration cues for large conflicts?
- Can data discriminate these computational strategies?

Unimodal duration discrimination


Unimodal Auditory
Signal
Noise
Test
(duration
varied)
Standard
(500ms)


Unimodal Visual
Test interval
(duration varied)
Standard
(500ms)
First
or Second
Longer?

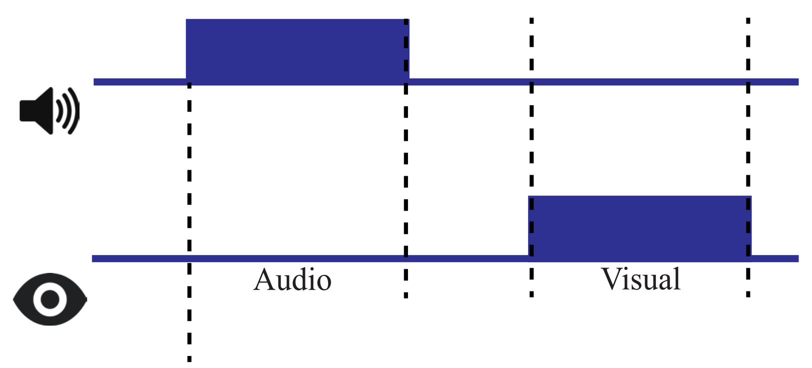

p ("Test longer")
Test duration (ms)
Visual
Low-noise auditory
High-noise auditory

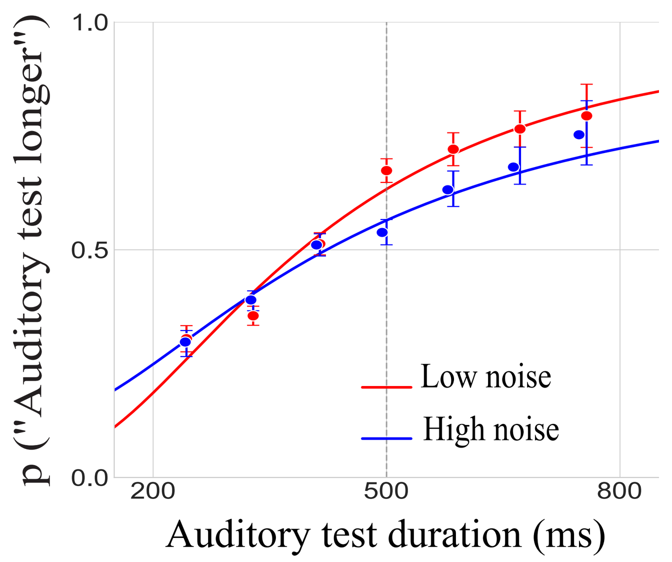

Weber fractions

**
n.s.
Low-noise
auditory
Visual
High-noise
auditory

Visual duration discrimination thresholds fell between low- and high-noise auditory thresholds.

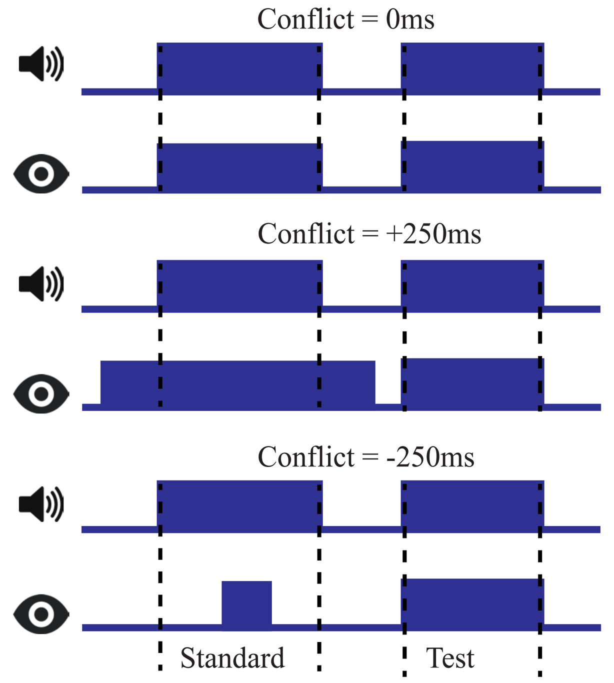
Cross-modal discrimination

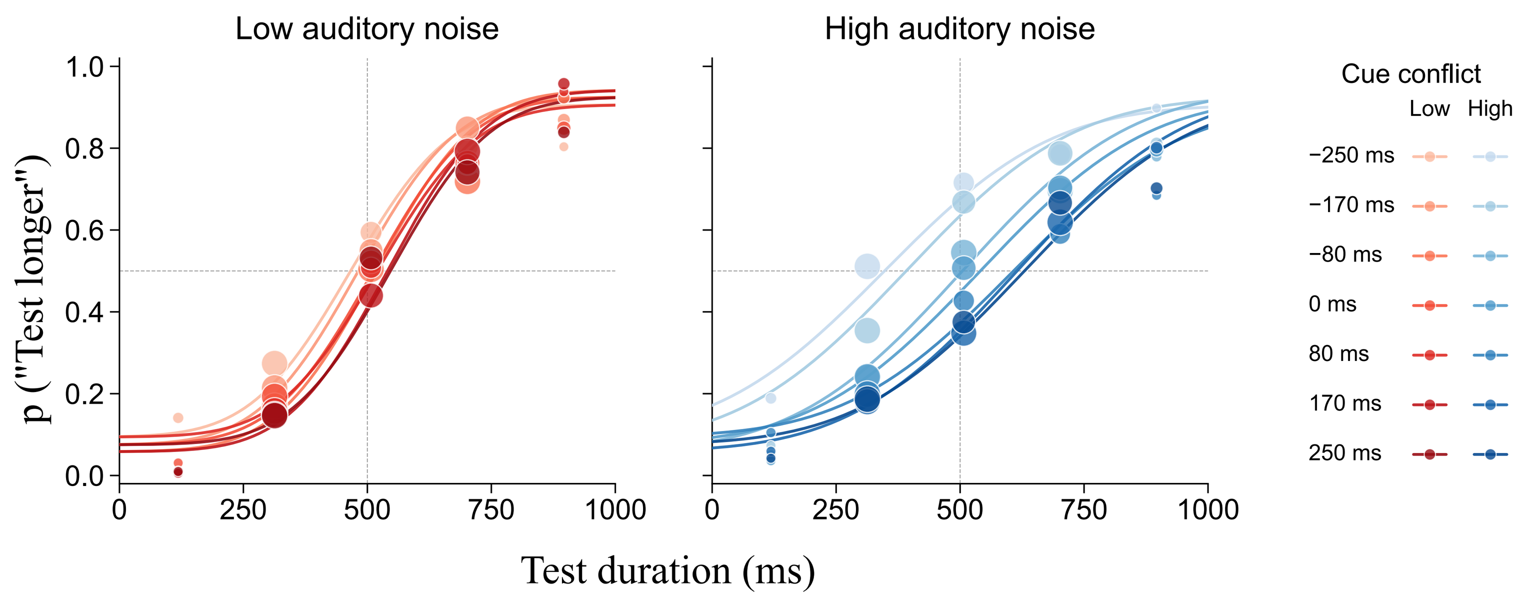

Audio
Visual


p ("Auditory test longer")
Auditory test duration (ms)
Low noise
High noise

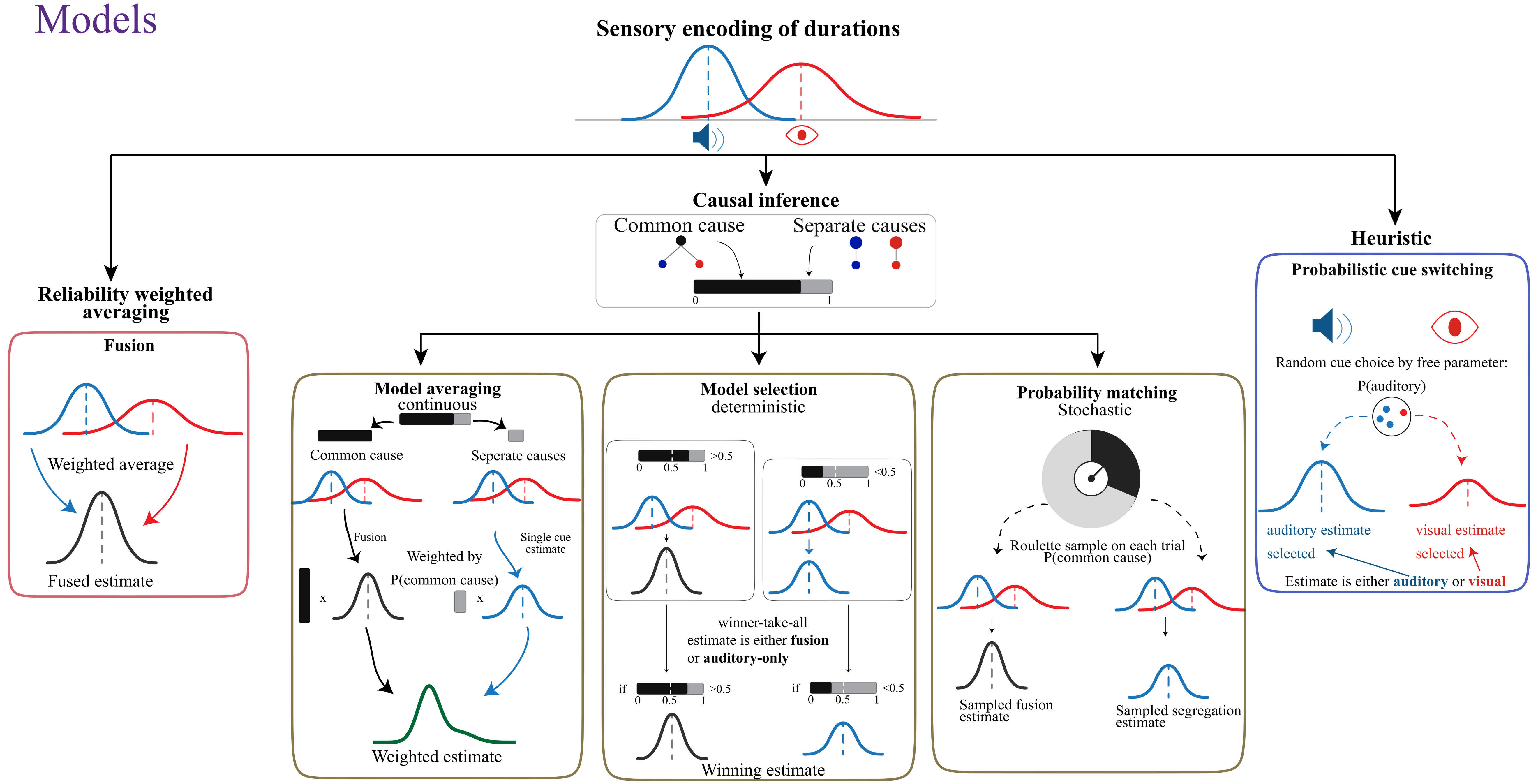
Auditory durations were perceived as longer than visual durations.

Bimodal duration discrimination


Conflict = 0ms
Conflict = +250ms
Conflict = -250ms
Standard
Test

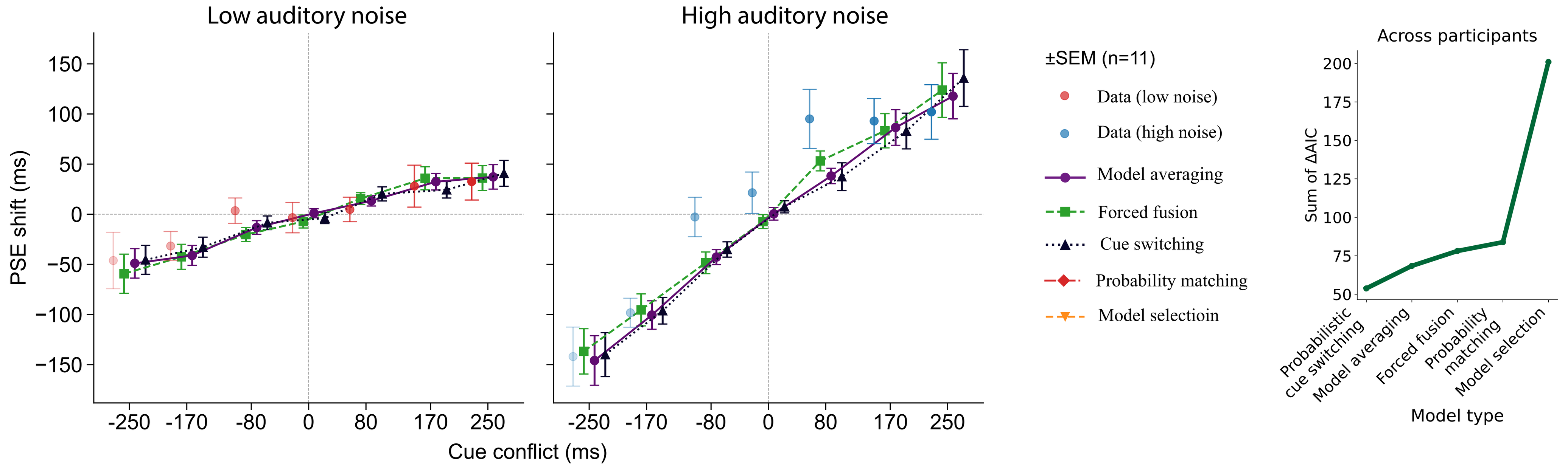

Low auditory noise
High auditory noise
p ("Test longer")
Test duration (ms)
Cue conflict
Low
High
−250 ms
−170 ms
−80 ms
0 ms
80 ms
170 ms
250 ms

Models

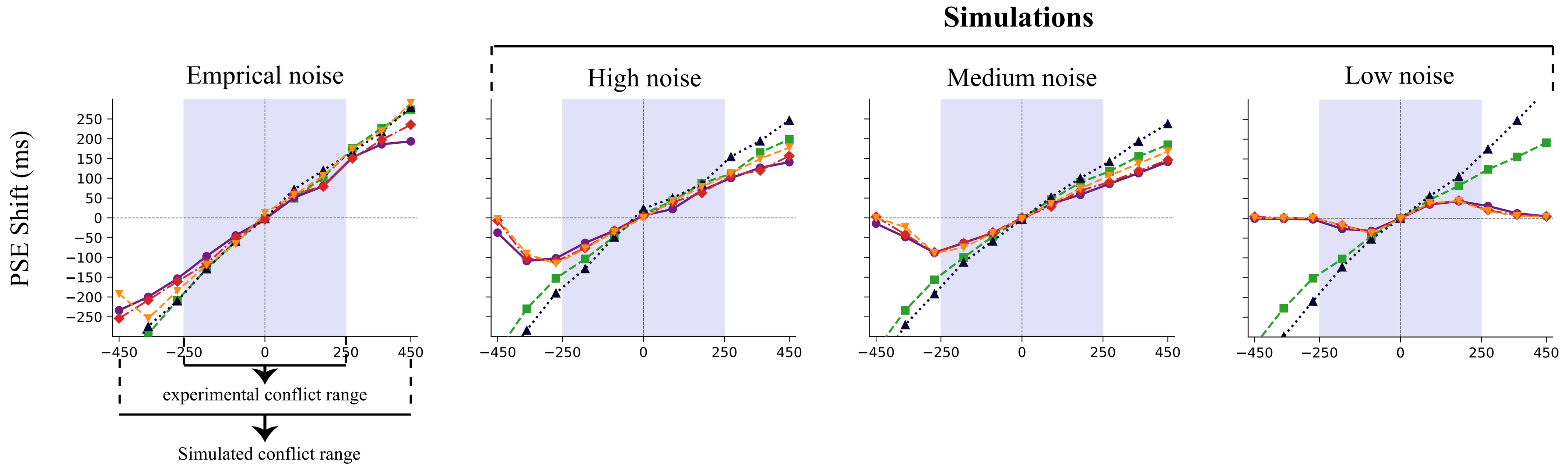

Sensory encoding of durations
Causal inference
Common cause
Separate causes
Reliability weighted
averaging
Fusion
Weighted average
Fused estimate
Model averaging
continuous
Common cause
Seperate causes
Fusion
Weighted by
P(common cause)
Single cue
estimate
Weighted estimate
Model selection
deterministic
winner-take-all
estimate is either fusion
or auditory-only
Winning estimate
Probability matching
Stochastic
Roulette sample on each trial
P(common cause)
Sampled fusion
estimate
Sampled segregation
estimate
Heuristic
Probabilistic cue switching
Random cue choice by free parameter:
P(auditory)
auditory estimate
selected
visual estimate
selected
Estimate is either auditory or visual

Models cannot be discriminated by our data

- All models produce very similar fits
- Cue-switching fits best for most participants, but other models fit nearly as well


Low auditory noise
High auditory noise
PSE shift (ms)
Cue conflict (ms)
±SEM (n=11)
Data (low noise)
Data (high noise)
Model averaging
Forced fusion
Cue switching
Probability matching
Model selectioin
Across participants
Sum of ΔAIC
Probabilistic
cue switching
Model averaging
Forced fusion
Probability
matching
Model selection
Model type

Model discriminability depends on sensory noise


Simulations
Emprical noise
High noise
Medium noise
Low noise
PSE Shift (ms)
experimental conflict range
Simulated conflict range

- Models could become discriminable under lower sensory noise than measured in the experiment and under an unreasonably wide range of cross-modal conflicts.

Interpretation

- Models can be discriminated in principle, but only with lower sensory noise than we found
- Participants have a consistent bias: auditory stimuli are perceived as longer
- The integration models produced highly similar behavioral predictions

1:Hartcher-O'Brien, J., Di Luca, M., & Ernst, M. O. (2014). The duration of uncertain times: audiovisual information about intervals is integrated in a statistically optimal fashion. PloS one, 9(3), e89339.
2:Körding, K. P., Beierholm, U., Ma, W. J., Quartz, S., Tenenbaum, J. B., & Shams, L. (2007). Causal inference in multisensory perception. PLoS one, 2(9), e943.

NIH EY08266, NYUAD Center for Brain and Health, funded by Tamkeen under NYU Abu Dhabi Research Institute award CG012